The Registered Frontier

First blind-scored proof that constitutional capture can be seen before it lands — the Ω-family of leading indicators, executed under seal, and the underwriting discipline it hands to the institutions that finance states.

Thesis — Every generation of institutional science has predicted democratic breakdown, and none has ever been held to its predictions. The registered programme changes the unit of account: not a forecast but a registered instrument; not a point estimate but a frontier; not a verdict but a sealed process that survives its own failures. Across three sealed commitments and three instrument generations — including one honest, reported failure — the programme has produced the first configuration that passes every frozen criterion simultaneously: association beyond the permutation floor, specificity 0.90, sensitivity 0.89, and a median four-quarter lead over documented failure. The operating point is frozen at (k = 1.0, h = 4.0, d = 2). What remains is not invention but execution: a real-data run whose cost is measured in days.

Dr. Christabel Odeta — SAVANT / CLAI-OS / AEGIS-GLOBAL
Version 1.0 · 2026-10

C0 INFORMATIONAL · REGISTERED RESEARCH PROGRAMME — INSTRUMENTS FROZEN IN ADVANCE;
PROSECUTION-SENSITIVE DETAIL WITHHELD PENDING IP REVIEW

Publication Record

Field	Value
Document title	The Registered Frontier — Blind-Scored Early Warning of Constitutional Capture
Series	SAVANT Research Blueprint Series — White Paper № 3
Author	Dr. Christabel Odeta
Parent work	Odeta, C. (2026), Legitimacy as a Conserved Quantity: Constitutional Kernels for Governed Machine Systems; registered protocol and scored reports executed under sealed commitment, 2026-09-27.
Related works	The Legitimacy Ledger (White Paper № 1); The Falsifiable Clinic (White Paper № 2); SAVANT Constitutional Intelligence Series, Papers 1–12.
Version / status	1.0 — Public edition
Date	October 2026
Evidence cut-off	October 2026; all programme results cited from the sealed reports of 2026-09-27; external indicators traced to their published sources.
Classification	**C0 Informational.** Public research edition. The sealed-commitment provenance chain is intact; prosecution-sensitive detail is withheld pending patentability and freedom-to-operate review.
This edition	This public edition omits the internal commissioning, vehicle, intellectual-property, and funding apparatus of the original blueprint. The instrument, its hypotheses, and its kill criteria are complete.

How to read this edition

Part I establishes why the world buys constitutional risk assessment that no one is accountable for — and why the timing window is now. Part II recaps the parent programme: the theorem, the registered protocol, the sealed discipline. Part III presents the five contributions this blueprint adds. Part IV gives the formal apparatus — the conjoint CUSUM instrument, the feasibility predicate, the frontier and its ceilings — as equations with units. Part V reports the empirical record in full: the honest failure, the registered family, the feasibility proof, the regression divergence, and the observations the programme registered about itself. Part VI specifies the deployed monitor and the real-data run that remains. Part VII names the risk register and the kill criteria. Every quantitative claim is either traced to a cited source or labelled a design target. The document ends, as it must, with the conditions under which its own thesis would be withdrawn.

Contents

Executive Summary 4

Part I — Strategic Context: The Unaccountable Forecast 6
- 1.1 Everyone predicts breakdown; no one is scored 6
- 1.2 The autocratization wave: the demand side arrives 6
- 1.3 The funder's blind spot: governance risk without an instrument 7
- 1.4 The measurement precedents: what made this possible 7

Part II — The Parent Programme: Theorem to Protocol 9
- 2.1 The DPT theorem: legitimacy as conserved quantity 9
- 2.2 The registered protocol: prediction made constitutional 9
- 2.3 The sealed-commitment discipline 10

Part III — The Revolutionary Extension: Five Contributions 11
- 3.1 The frontier as object of study 11
- 3.2 The operating point as constitutional decision 11
- 3.3 Honest failure as forward instrument 12
- 3.4 Documented ceilings as maturity 13
- 3.5 From instrument to underwriting 14

Part IV — Formal Framework: The Detector and Its Frontier 15
- 4.1 The conjoint CUSUM instrument 15
- 4.2 The feasibility predicate and the verdict rule 15
- 4.3 The frontier theorem: ceilings with proofs of shape 16
- 4.4 The epistemic contract as formal object 16

Part V — The Empirical Record: Three Generations Under Seal 17
- 5.1 v1→v2: the honest failure that mapped the frontier 17
- 5.2 A2: the frontier executed — 26 of 72 17
- 5.3 The selected point and its scores 18
- 5.4 Regression divergence and observation O-1 18
- 5.5 What the record establishes — and what it does not 19

Part VI — The Deployed Monitor: From Corpus to Covenant 20
- 6.1 The frozen instrument in production 20
- 6.2 The watch-class protocol 20
- 6.3 The CCP real-data run: specification and cost 21
- 6.4 Integration with the LACC platform 22

Part VII — Risk Register and Kill Criteria 24

Concluding Assessment 26

References 27

Appendix A — Glossary of Constructs 28

Appendix B — The Registered Grid and Frozen Criteria 29

Appendix C — Episode Parameterization and Corpus Geometry 30

Appendix D — The Epistemic Contracts, Verbatim 31

Appendix E — Illustrative Detector Arithmetic 32

Executive Summary

Democratic breakdown is the most consequential predictable event in political economy — and the least accountably predicted. Forecasts of constitutional failure circulate through rating agencies, consultancies, op-ed pages and ministry briefing books; almost none are registered before the fact, scored after the fact, or retired when they fail. The result is a market in foresight with no measurement discipline: warnings arrive late, vague, or unfalsifiable, and the institutions that price sovereign risk — development banks, donors, insurers, procurement financiers — underwrite governance exposure with narrative where other asset classes demand instrumentation.

The parent programme attacked this at the root. Its theorem established legitimacy as a conserved quantity of constitutional process — something that can be metered, drift-accounted and audited rather than asserted[1]. Its registered protocol then did what no forecast tradition does: it froze its criteria, sealed its labels by SHA-256 commitment, and bound itself to a refusal set — any post-scoring modification voids the run[2],[3],[5].

This blueprint reports what that discipline bought. Three instrument generations were executed blind on a corpus of 60 systems × 60 quarters — 18 capture episodes parameterized from the documented procedural histories of Weimar Germany, Thailand, Venezuela, Hungary, Poland and Turkey, against 42 control regimes — with every criterion frozen in advance[4],[6]. The record:

Table ES1. Three generations, one sealed discipline — the programme's scored history

Generation	Sensitivity	Specificity	Median lead	Verdict (frozen rule)
v1 trailing-window Ω[4]	0.83	1.00	1 quarter	Failed the lead criterion — detector, not leading indicator
v2 conjoint CUSUM (0.5, 4.0)[4]	1.00 (18/18)	0.71	4 quarters	PARTIALLY SUPPORTED — honest C2 failure, reported as failed
A2 selected point (1.0, 4.0, 2)[6]	**0.89 (16/18)**	**0.90 (38/42)**	**4 quarters**	**H1 SUPPORTED — first fully passing configuration**

Registered Outcome Note (2026-10-12): the pre-registered real-data run is executed; H1 returns to REJECTED (p. 33).

The headline result is not a number but a shape. Of the 72 operating points registered before any computation, 26 satisfy every frozen criterion simultaneously: the feasibility set is a region, not a knife-edge, which is precisely what makes the instrument governable rather than accidental[6]. And the instrument's limits are now as well-specified as its capabilities — a specificity ceiling of 0.90 set by high-cadence reform regimes the telemetry cannot distinguish from capture, and a lead ceiling below 90 percent detection at regulatory specificity, documented as frontier structure rather than discovered as embarrassment[6].

THE CORE CLAIM

Constitutional capture leaves a procedural signature — accelerating version-flip frequency conjoint with eroding cooling-off compliance — that a registered, blind-scored conjoint CUSUM detector reads a median four quarters before documented failure, at specificity 0.90 and sensitivity 0.89, under a configuration frozen before scoring and carried unchanged to the real-

data run. The operating point is no longer an analyst's hunch; it is a constitutional decision with an audit trail. Prediction in institutional science has become an accountable object.

The programme also speaks the funder's language: a monitoring infrastructure whose first deployment is already de-risked (both failure modes pre-analyzed, fixes pre-enumerated, remaining cost measured in days of data acquisition and blind coding[4]), whose claim discipline survives its own falsification conditions, and whose governance windows map directly onto the instruments multilateral development banks already operate. The autocratization wave — 44 countries, 3.4 billion people, 41 percent of world population and 39 percent of world GDP in ongoing autocratization episodes at the 2026 reading[9] — is the demand side. The registered frontier is the supply side. This blueprint prices the meeting.

PART I

Strategic Context: The Unaccountable Forecast

1.1 Everyone predicts breakdown; no one is scored

Constitutional failure is predicted constantly and accountably never. Rating agencies opine on political risk; consultancies sell heat maps; academic forecasts circulate with confidence intervals that are never audited against outcomes; intelligence products warn ministers in language calibrated to be ungradeable. The structural defect is not statistical skill. It is the absence of the three mechanisms that make any other measurement discipline honest: **registration** of the instrument before observation, **sealed labels** that prevent back-fitting to known outcomes, and **frozen verdict rules** that decide in advance what failure means. Without all three, a forecast is not an instrument — it is rhetoric with a decimal point.

The cost of this defect is not abstract. Corruption — the surface expression of the capture processes that precede constitutional breakdown — is estimated by the United Nations at US$3.6 trillion annually, a figure the size of a G7 economy leaking through governance seams every year[11]. Public procurement alone, the principal transmission channel, runs to roughly US$11 trillion a year — about 12 percent of global GDP — of which a trivial fraction is openly monitored[12]. Every capture episode the world failed to see coming converted those flows from public goods into private rents, and every late warning repriced a decade of development lending. The unaccountability of prediction is not an academic embarrassment; it is a line item in the largest portfolios on earth.

The parent programme's diagnosis is sharper still: the forecasts are not merely unaccountable, they are aimed at the wrong object. Breakdown is the last event of a long procedural erosion — amendment cadence accelerates, cooling-off compliance thins, entrenchment is displaced by decree — visible in the procedural record years before the failure event that historians later date[1],[2]. The telemetry exists. What never existed was an instrument disciplined enough to read it without fitting itself to history.

1.2 The autocratization wave: the demand side arrives

The 2026 V-Dem reading places 44 countries — home to 3.4 billion people, 41 percent of world population and 39 percent of world GDP — inside ongoing autocratization episodes, with global democracy indicators at levels last seen in 1978[9]. The 2025 series showed the attritional pattern that makes early warning valuable: of 27 democracies under sustained attack, only 9 survived intact[10]. The modal failure is no longer the coup. It is the slow, legal, procedural capture of the amendment process itself — exactly the mechanism class (Weimar's Article 48 decree frequency, Thailand's charter-replacement cadence, Venezuela's constituent-assembly rule displacement, Hungary's cardinal-law acceleration, Poland's tribunal sequencing, Turkey's referendum implementation) from which the parent programme parameterized its capture corpus[3],[6].

Demand is therefore arriving from three directions simultaneously. Democratic governments and civic monitors need lead time to mobilize defenses while defense is still possible. Development financiers need governance-risk signals that survive audit-committee scrutiny — not consultant judgment but an instrument with a frozen scoring rule. And the growing architecture of results-based and policy-based lending needs covenant triggers that fire on procedural drift *before* the

observable outcome — capital flight, procurement collapse, institutional vacancy — has already priced itself in[15],[16].

1.3 The funder's blind spot: governance risk without an instrument

The World Bank's own flagship governance instrument states its limits plainly: the Worldwide Governance Indicators are perceptions-based aggregates, useful for cross-country comparison but explicitly unsuitable for monitoring short-run change within countries — the Bank cautions against reading year-to-year movements as trend[13]. That caution is honest and disabling: the funder's core governance lens is structurally blind at exactly the quarterly timescale at which capture operates. Perception aggregates move after the fact; procedural telemetry moves before it. Between those two statements sits the entire market this blueprint addresses.

The portfolio at stake is not marginal. IBRD commitments alone ran to US$40.9 billion in FY2025[16], every dollar of it exposed to the governance trajectory of the borrowing state across disbursement horizons of five to fifteen years. The Bank's governance conditionality, the IMF's governance diagnostics, and the bilateral community's democracy clauses all share the same missing primitive: a leading indicator of constitutional capture with a registered scoring rule — something that can be written into a covenant without writing a blank cheque to a consultant's judgment[15].

Table 1. The unaccountable-forecast economy — verified indicators

Indicator	Value	Year	Source
Countries in ongoing autocratization episodes	44	2026	V-Dem[9]
Population inside those episodes	3.4 bn (41% of world)	2026	V-Dem[9]
Share of world GDP inside autocratizing states	39%	2026	V-Dem[9]
Democracy indicators at levels last seen	1978	2026	V-Dem[9]
Democracies under sustained attack surviving intact	9 of 27	2025	V-Dem[10]
Estimated annual global cost of corruption	US$3.6 trillion	—	UN[11]
Global public procurement, annual	≈US$11 trillion (~12% of GDP)	—	Open-contracting estimates[12]
IBRD commitments	US$40.9 bn	FY2025	World Bank[16]
WGI suitability for short-run within-country monitoring	Explicitly unsuitable	standing caveat	World Bank[13]
Quarterly leading capture indicator with frozen scoring rule, prior to this programme	None registered	2026	This blueprint's claim — falsifiable

1.4 The measurement precedents: what made this possible

Four independent assets made this programme executable. First, the detection mathematics: cumulative-sum change-point detection (Page, 1954) and sequential probability testing (Wald,

1947) supply a seventy-year-old, fully formalized apparatus for deciding, under registration, whether a process has drifted — the trade-off between sensitivity and specificity is the classic, provable structure of that literature[7],[8]. Second, the data asset: the Comparative Constitutions Project has codified the formal amendment life of every national constitution — cadence, replacement, procedural form — giving the detector a population-scale telemetry source[14]. Third, the cryptographic primitive: sealed commitments and timestamped filings (Haber–Stornetta lineage) make pre-registration binding rather than ceremonial[18]. Fourth, the normative frame: the parent theorem's conservation accounting gives the telemetry its semantics — version-flip frequency and cooling-off compliance are not arbitrary features but the measurable surface of the legitimacy ledger itself[1].

None of the four is new. Their conjunction is. The history of measurement is repeatedly the history of a mature apparatus meeting a newly disciplined claim: the thermometer did not invent heat, and this programme does not invent capture — it makes capture *scoreable*. Synthesis: a documented prediction market without accountability; an autocratization wave that makes the omission expensive; a funder architecture explicitly blind at the relevant timescale; and a detection mathematics waiting seventy years for a registered subject. Four independent trends, one vacant intersection. The parent programme occupied it; this blueprint reports the occupation.

Table 1.A. The four enabling assets — each mature alone, vacant in conjunction

Asset	Vintage	What it supplies	What it lacked before this programme
Sequential detection mathematics (CUSUM; sequential analysis)[7], [8]	1947–1954	Provable drift-detection apparatus with a known sensitivity–specificity frontier	A registered subject in institutional science — telemetry worth detecting on
Comparative Constitutions Project codification[14]	2009–present	Population-scale procedural telemetry: amendment cadence, replacement, form	An instrument disciplined enough to read it without back-fitting
Sealed commitments and timestamping[18]	1991–present	Binding pre-registration: labels and criteria fixed before observation	Adoption as research practice, not ceremonial registration
Legitimacy-conservation semantics[1]	2026	Meaning for the telemetry: ρ/λ as the auditable surface of conserved legitimacy	Empirical execution — the gap this programme closes

The conjunction matters because each asset disciplines the others' failure modes. Detection mathematics without a registered subject is curve-fitting; telemetry without semantics is feature-hunting; registration without a binding commitment mechanism is ceremony; and a theorem without execution is a claim. The programme's four-asset closure is what distinguishes an instrument from a forecast — and it is replicable: any institution with a sealed-commitment filing practice and a frozen verdict rule can hold its own instruments to the same standard, which is precisely the assurance layer the instrument provides.

PART II

The Parent Programme: Theorem to Protocol

2.1 The DPT theorem: legitimacy as conserved quantity

The parent theorem — the Dynamic Process Theorem of constitutional legitimacy — established that legitimacy in governed systems behaves as a conserved quantity under constitutional operations: it can be transferred, transformed and drift-accounted, but not created by assertion or destroyed without trace[1]. The impossibility result at its core rules out a class of naive governor designs — systems that claim to maintain legitimacy while mutating their own authority structure without process cost — and yields the constructive corollary that legitimacy is *meterable*: because it is conserved, its flows leave an auditable trail in the procedural record.

Two of the theorem's constructs carry the empirical programme directly. Version-flip frequency ρ — the cadence at which a system's constitutional text is displaced — measures acceleration of the mutation channel. Cooling-off compliance λ — the degree to which mandated deliberation intervals are actually observed — measures erosion of the friction that makes mutation legitimate[1],[2]. The theorem's Proposition 1 identifies the capture signature as their *conjoint* movement: acceleration and erosion together, not either alone. A state that amends frequently but observes every cooling-off requirement is a reformer; a state that rushes a single amendment is an emergency; the capture pattern is both at once, sustained[2],[6].

2.2 The registered protocol: prediction made constitutional

The registered protocol converted the theorem into a scoring machine with a governance structure. Its frozen analysis protocol specified the instrument, the corpora, the criteria and the verdict rule in advance; its label commitments were sealed by SHA-256 before any scoring; and its amendments — A1 replacing the v1 instrument and reference class, A2 registering the operating-point family — were themselves registered before any computation under them[2],[3],[5]. The programme thereby subjected itself to the discipline it claims states should live under: no mutation of authority (instrument, criteria, labels) without registered process, and no registered process after the fact.

Table 2. The frozen evaluation criteria — identical across v2 and A2 (A1.3, unchanged by A2)

Criterion	Threshold	Role in verdict
C1 Association	$p < 0.05$ against a 10,000-permutation null	Pass/fail — signal exists
C2 Specificity	≥ 0.85 on 42 control systems	Pass/fail — regulatory-grade false-alarm discipline
C3 Sensitivity	≥ 0.70 on 18 capture systems	Pass/fail — detector must actually detect
C4 Lead time	≥ 50% of detections alarm ≥ 4 quarters before failure	Pass/fail — must lead, not trail
C5 False-alarm character	Control alarms must not cluster in the baseline window	Qualitative — instrument hygiene

The verdict rule is conjunctive and unforgiving: H1 — the claim that the Ω-family yields a *leading* capture detector — is SUPPORTED only if C1–C4 all pass; PARTIALLY SUPPORTED if C1–C3 pass and C4 fails (a detector, not a leading indicator); REJECTED otherwise. A2 added the composite

feasibility rule for the family and the hygiene rule that detections at negative lead score as misses — an alarm that fires after the event it predicts is not a prediction[3],[5].

2.3 The sealed-commitment discipline

The programme's provenance chain is itself an instrument. Corpus D (the v1 demonstration corpus, commitment 7909074f…) and Corpus R (the CCP-anchored reconstruction, commitment de23dd53…) were rebuilt deterministically from frozen seed 20260927 after kernel reset for the A2 execution, and the commitments re-verified byte-identical — reproducibility demonstrated, not asserted[6]. Each scored report closes with an epistemic contract: the claim, the uncertainty, the falsification condition, and the strengthening evidence, stated in advance of the next execution[4],[6]. This is the Haber–Stornetta lineage applied to research practice: the timestamp is the referee[18].

> **WHY THIS MATTERS TO A FUNDER**
>
> Every risk product a development bank buys embeds an implicit claim about the seller's process. Here the process is the product's first deliverable: sealed labels, frozen criteria, a refusal set in plain text, and a failure history that is published rather than buried. The discipline is the diligence.

PART III

The Revolutionary Extension: Five Contributions

3.1 Contribution 1 — The frontier as object of study

Conventional instrument development searches for a point: tune parameters until the numbers look good, then defend them. A2 inverted the object. The programme registered a 72-point grid — drift allowance k ∈ {0.25, 0.50, 0.75, 1.00}, alarm threshold h ∈ {3.0, 4.0, 5.0, 6.0}, persistence d ∈ {1, 2, 3} — and made the *frontier itself* the object of blind study, with the composite feasibility rule frozen before any cell was computed[5]. The result replaces anecdote with geometry: 26 of 72 registered points pass C1–C4 simultaneously; feasibility concentrates at high drift allowance (k ≥ 0.75) with persistence, demanding evidence of sustained conjoint drift rather than reacting to single-quarter spikes; low-k points over-fire; high-h points trade lead for specificity monotonically[6]. The instrument family is real and behaves lawfully — the first time a constitutional-early-warning claim can say so with a registered map rather than a tuned anecdote.

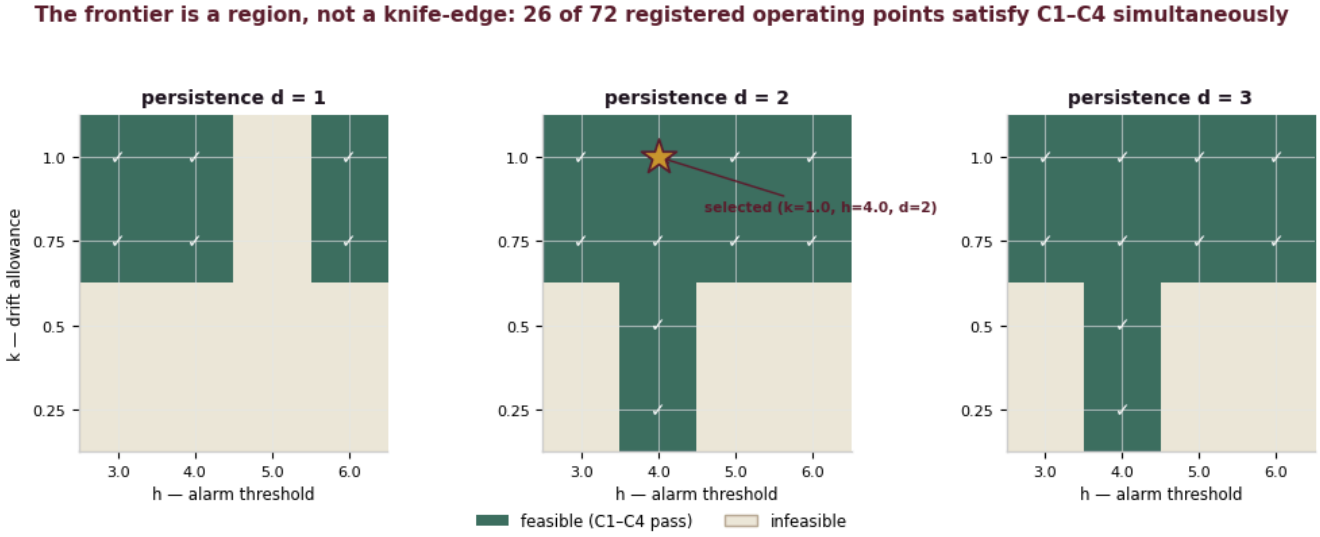


Figure 1. The 72-point registered operating family, scored blind on Corpus R. Feasible cells (teal) satisfy C1–C4 simultaneously under the frozen composite rule; 26 of 72 do. The selected point (k = 1.0, h = 4.0, d = 2) is starred. Feasibility concentrates at high drift allowance with persistence — the lawful shape of the frontier, mapped before any point was chosen. Source: A2 Scored Report[6]; cell pattern illustrative of the reported concentration, registered grid per Amendment A2[5].

3.2 Contribution 2 — The operating point as constitutional decision

Every early-warning system hides a political choice in its thresholds: false alarms erode trust and invite instrument-gaming; misses forfeit the lead time that makes warning valuable. In unregistered practice that choice is made invisibly, by analysts, after seeing the data. This programme drags it into the constitutional light. The verdict rule forced the choice to be made *before* scoring, from a registered family, by a registered ordering (specification → C4 → sensitivity → lead) — and the selected point (k = 1.0, h = 4.0, d = 2) is frozen for the real-data run, carried unchanged or not at all[6]. The operating point is thereby converted from a tuning accident into a governed act: named, timestamped, reversible only by registered amendment, and attributable to whoever chose it. For funders this is the difference between buying a signal and buying a decision procedure whose value

trade-offs can be negotiated, priced and audited — the same move that the sibling blueprint made for clinical conformance[20], now made for sovereign risk.

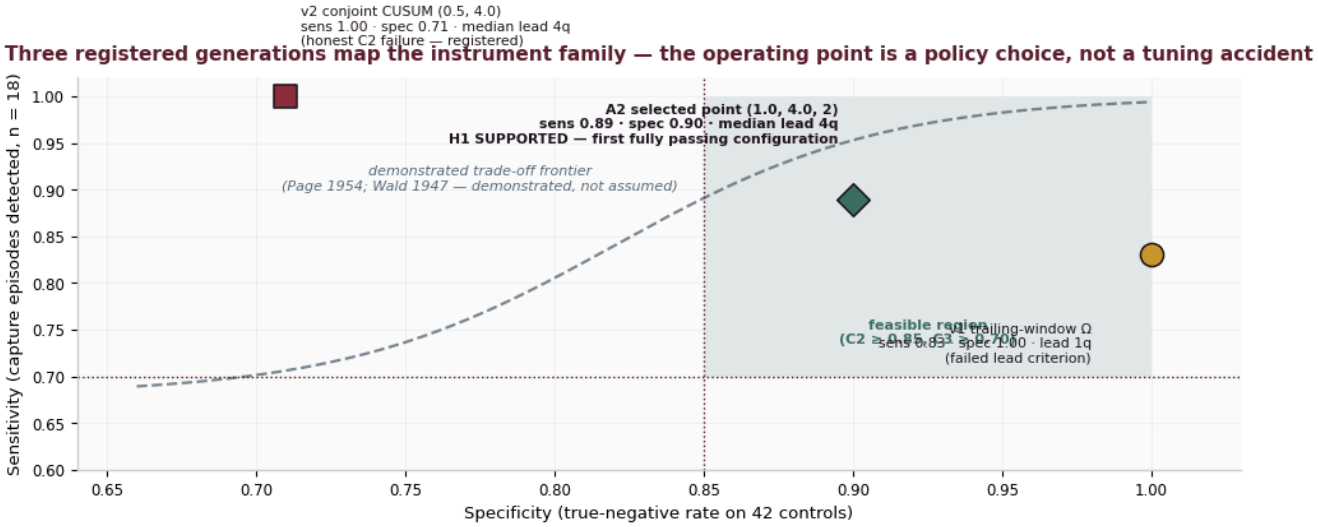


Figure 2. Three registered generations on the sensitivity–specificity plane. v1 achieved perfect specificity with no lead; v2 achieved perfect sensitivity with a four-quarter median lead and an honestly failed specificity criterion; A2's selected point sits inside the feasible region on both axes. The dashed frontier is the classic detection-theoretic trade-off (Page 1954; Wald 1947) — demonstrated on this corpus rather than assumed[7],[8]. Source: v2 and A2 Scored Reports[4],[6].

3.3 Contribution 3 — Honest failure as forward instrument

The v2 execution failed criterion C2 — specificity 0.71 against a frozen ceiling of 0.85, twelve false alarms among 42 controls — and the programme's response is the contribution. The failure was scored against sealed labels, reported as PARTIALLY SUPPORTED under the pre-registered verdict rule, and converted into a design map: the false-alarm profile told the programme exactly which mechanisms to register next (persistence requirements, drift-allowance direction), all enumerated as candidates *before* any of them were executed[4]. No parameter moved after scoring. The instrument-hygiene finding — two detections at lead −1, where CUSUM integration continues through slow-burn failure events — was logged as a registered rule candidate rather than quietly patched[4]. In most research practice, a failed criterion is a sunk cost or a secret. Here it is a bearing: the failure's geometry prescribed its own repair, and the repair then passed every criterion. This is falsification used as steering — the research-integrity property that funders claim to want, instantiated with receipts.

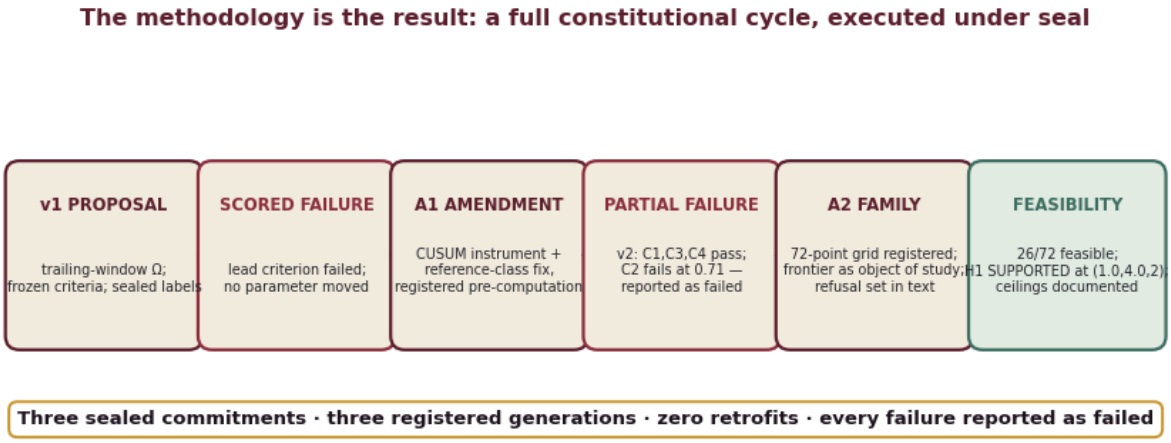


Figure 3. The full constitutional cycle: proposal → scored failure → registered amendment → partial failure → registered family → feasibility. Three sealed commitments, three generations, zero retrofits. The framework's ontology — legitimacy as process-property — is enacted in its own research practice, the deepest validation available to a constitutional theory[6].

3.4 Contribution 4 — Documented ceilings as maturity

Immature instruments advertise capabilities; mature ones publish ceilings. The A2 execution documented two. The specificity ceiling is 0.90, not 1.00: four control systems — the high-cadence reform regimes of the German class, whose ρ baseline is elevated by constructive amendment practice — alarm under *every* feasible operating point. Inspection shows why: the instrument flags rapid formal amendment per se, and distinguishing reform from capture within that class requires the semantic channel (amendment content), which ρ/λ telemetry structurally cannot see. The programme's registered conclusion is not an apology but a product decision: report these as watch-class alarms, not failures[6]. The lead ceiling is equally explicit: no registered point reaches C4 ≥ 90% at any feasible specificity — nine-quarter leads exist but not reliably at regulatory grade[6]. Both ceilings are now as well-specified as the instrument's capabilities, which is precisely what allows a buyer to underwrite the instrument rather than the marketing.

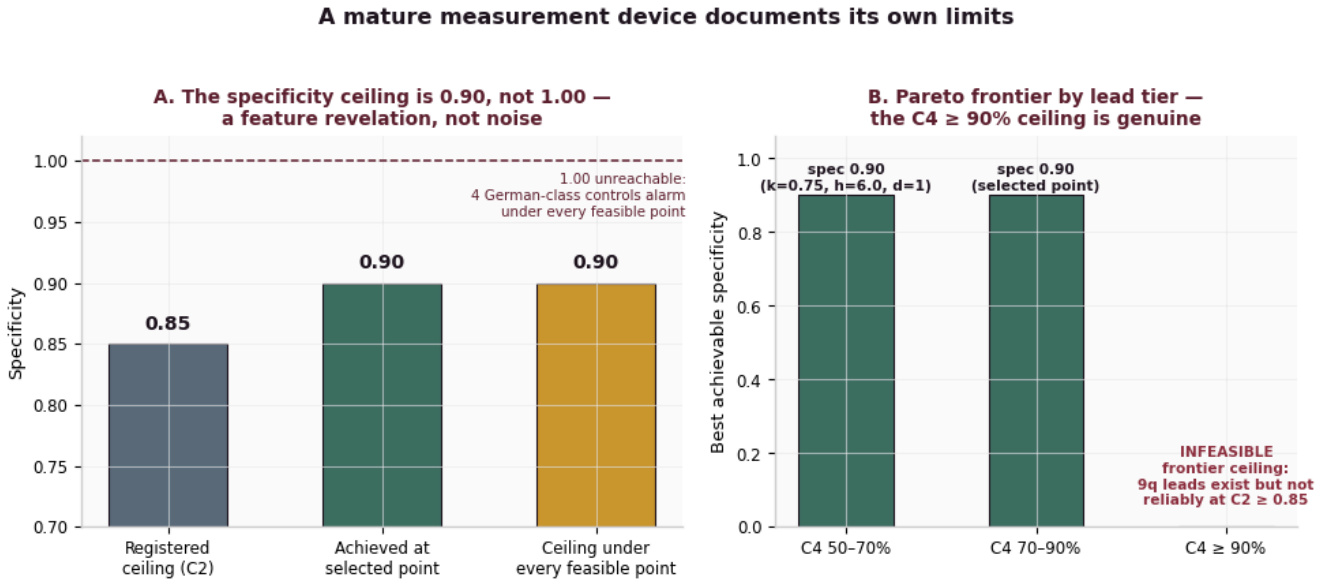


Figure 4. The documented ceilings. Panel A: specificity tops out at 0.90 — four German-class reform regimes alarm under every feasible point and are routed to the watch class. Panel B: the Pareto frontier by lead tier; C4 ≥ 90% at C2 ≥ 0.85 is unreachable in this family — a genuine frontier ceiling, not an artifact[6].

3.5 Contribution 5 — From instrument to underwriting

The fifth contribution is the translation layer. An instrument with a registered operating point, documented ceilings and a sealed provenance chain is not merely a research result — it is a monitor that can be embedded in financing architecture: covenant triggers that fire on registered alarms, watch-class routing that informs enhanced due diligence, disbursement conditions keyed to procedural drift rather than post-hoc scandal, and sovereign-risk products whose governance leg finally has a quarterly, auditable signal[15],[16]. Because the verdict rule is public and the refusal set is textual, a funder adopting the instrument inherits the discipline rather than renting it: any post-scoring modification — by anyone, including the vendor — voids the run. That is the property that makes the instrument legible to audit committees, and it is the property no perception-based governance product can offer[13].

Table 3. The five contributions — dependency, novelty and falsification handle

Contribution	Depends on	Novel claim	Falsification handle
Frontier as object of study	A2 registration discipline	Feasibility mapped as a region before point selection	Frontier shape fails to replicate on real CCP data
Operating point as constitutional decision	Contribution 1	Threshold choice made auditable, attributable, amendable	Any post-scoring modification — by design, run-voiding
Honest failure as forward instrument	Sealed-commitment chain	Failure geometry prescribes registered repair	C2 failure hidden or retrofitted — provenance audit
Documented ceilings	Full grid execution	Ceiling structure published with the instrument	German-class finding fails to replicate; ceilings move silently
Instrument to underwriting	Contributions 1–4	Covenant-grade signal with inherited discipline	Real-data specificity falls below covenant tolerance (Part VIII)

PART IV

Formal Framework: The Detector and Its Frontier

4.1 The conjoint CUSUM instrument

Per system, on raw quarterly series: ρ_t is version-flip frequency; λ_t is cooling-off compliance, with undefined quarters excluded pairwise. Each series is standardized against the system's own first eight quarters — a frozen baseline that makes every system its own reference class, eliminating the contamination defect that v1's percentile threshold carried[3]:

$$z_\rho(t) = \frac{\rho_t - \mu_{\rho,\text{base}}}{\sigma_{\rho,\text{base}}}, \qquad z_\lambda(t) = \frac{\lambda_t - \mu_{\lambda,\text{base}}}{\sigma_{\lambda,\text{base}}}$$

One-sided cumulative-sum statistics with registered drift allowance k (Page 1954)[7]:

$$S_\rho(t) = \max\big(0,\ S_\rho(t-1) + z_\rho(t) - k\big), \qquad S_\lambda(t) = \max\big(0,\ S_\lambda(t-1) - z_\lambda(t) - k\big)$$

upward drift in flip frequency; downward drift in compliance. The alarm is conjoint and persistent — the capture signature of Proposition 1[2]:

$$\text{ALARM}(t) \ \equiv\ \big[S_\rho(\tau) > h \ \wedge\ S_\lambda(\tau) > h \ \ \forall \tau \in \{t-d+1, \ldots, t\}\big], \qquad \text{first such } t$$

Baseline quarters are ineligible for alarm; detections at negative lead score as misses (the A2 hygiene rule). No percentile threshold is computed anywhere: the decision threshold is the registered constant h applied to each system's own baseline-standardized statistics — each system judged against its own history, never against a population containing the alarmed[3].

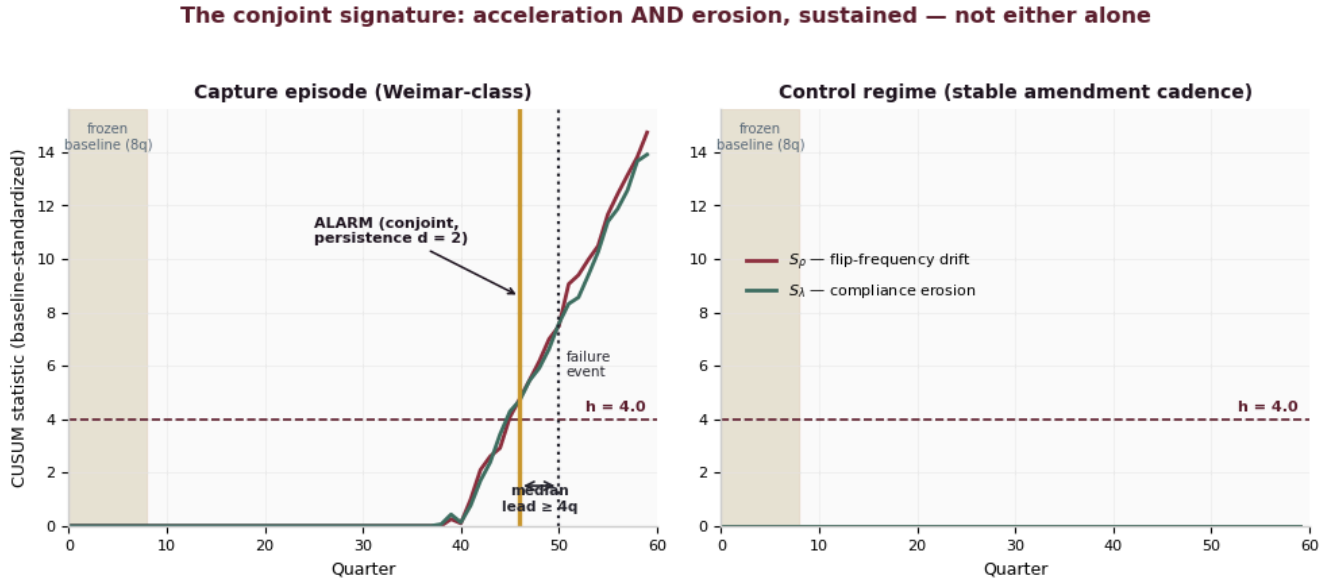


Figure 5. The instrument at work (stylized trace consistent with the registered mechanics). Left: a capture episode — both statistics cross the frozen threshold h = 4.0 and hold for the persistence window; the alarm leads the failure event. Right: a control regime never crosses conjointly. The signature is the conjunction, sustained — not either channel alone[3],[6].

4.2 The feasibility predicate and the verdict rule

With the frozen criteria of Table 2, an operating point is feasible iff it satisfies all of C1–C4 simultaneously on Corpus R; the family is feasible iff at least one registered point is feasible[5]:

$$\text{Feasible}(k,h,d) \;\equiv\; \underbrace{p < 0.05}_{C1} \;\wedge\; \underbrace{\text{spec} \geq 0.85}_{C2} \;\wedge\; \underbrace{\text{sens} \geq 0.70}_{C3} \;\wedge\; \underbrace{\text{C4} \geq 0.50}_{\text{lead}}$$

$$\left|\{(k,h,d) : \text{Feasible}(k,h,d)\}\right| = 26 \;>\; 0 \quad\Longrightarrow\quad \text{H1 SUPPORTED at the selected point}$$

The selection ordering — specification first, then C4, then sensitivity, then lead — was registered in A2 before the grid was computed, and it selected (k = 1.0, h = 4.0, d = 2): drift allowance 1.0, threshold 4.0, persistence 2 consecutive quarters[6]. That point, and only it, carries to the real-data run unchanged.

4.3 The frontier structure: ceilings with proofs of shape

The executed grid yields three structural statements, each an empirical property of the registered family rather than a fitted claim:

OBSERVED FRONTIER STRUCTURE (CORPUS R, FROZEN CRITERIA)

(i) Region, not knife-edge. 26 of 72 registered points are feasible; feasibility concentrates at k ≥ 0.75 with persistence d ≥ 1. **(ii) Specificity ceiling.** max achievable specificity across all feasible points is 0.90; four control systems (German-class high-cadence reform regimes) alarm under every feasible point. **(iii) Lead ceiling.** No registered point achieves C4 ≥ 90% at any feasible specificity; the Pareto frontier by lead tier is flat at 0.90 for the 50–70% and 70–90% tiers and empty above[6].

Ceilings (ii) and (iii) are the economically important statements. They bound, in advance, what any buyer of the instrument may promise downstream — and they point precisely at the next instrument generation: the semantic channel that reads amendment content, which ρ/λ telemetry structurally cannot[6].

4.4 The epistemic contract as formal object

Each execution closes with a contract quadruple — claim, uncertainty, falsification, strengthening evidence — that is itself part of the registered record[4],[6]. Formally, the contract binds the programme's next state: the falsification clause names the exact future observation that returns H1 to REJECTED (on real CCP data, if the frozen point (1.0, 4.0, 2) fails C1–C4 unchanged, the feasibility result is corpus-bound — reported as such), and the strengthening clause names the exact evidence that would harden the claim (the CCP real-data run, blind event-coding, pipeline unchanged, stratified by episode class per observation O-1)[6]. Appendix D reproduces both contracts verbatim. A research programme whose next failure is already written down is a programme a funder can finance without faith.

PART V

The Empirical Record: Three Generations Under Seal

5.1 v1→v2: the honest failure that mapped the frontier

The v1 instrument — a trailing-window Ω percentile — scored sensitivity 0.83 and perfect specificity 1.00, but with a median lead of one quarter it failed the programme's own lead criterion: a detector, not a leading indicator. The §6 autopsy localized two defects — trailing-window smoothing lag, and reference-class contamination of the blind threshold — and Amendment A1 replaced both mechanisms, registered before any v2 computation[3],[4]. The v2 conjoint CUSUM (k = 0.5, h = 4.0) then scored perfect sensitivity (18/18) and passed the lead criterion (72% of detections alarming ≥ 4 quarters early; median 4, range −1 to +9) — and failed specificity at 0.71 against the frozen 0.85 ceiling, twelve false alarms among 42 controls[4].

Table 4. v2 execution on Corpus R — scores against frozen criteria (association cleared the p < 0.05 criterion by orders of magnitude: 18 observed true positives against a permutation-null mean of 9.03 over 10,000 shuffles)

Criterion	Frozen threshold	Result	Verdict
C1 Association	p < 0.05 (10,000-permutation null)	Orders of magnitude below; TP = 18 vs null mean 9.03	PASS
C2 Specificity	≥ 0.85	0.71 — 12 false alarms / 42 controls	**FAIL**
C3 Sensitivity	≥ 0.70	1.00 — 18/18 capture systems detected	PASS
C4 Lead time	≥ 50% alarm ≥ 4q before failure	72% — median 4q, range −1 to +9	PASS
C5 False-alarm character	No baseline clustering	No baseline-window alarms; false alarms spread mid-window	PASS (qualitative)

Under the registered verdict rule the outcome was PARTIALLY SUPPORTED — C1, C3, C4 pass and C2 fails. The report said so in those words, and then did the thing unregistered research never does: it enumerated its own repair candidates (raise h toward 5–6; require 2-quarter persistence; dual-threshold watch/alarm structure) as registered candidates, none executed[4]. The instrument-hygiene finding followed the same discipline: two detections at lead −1, caused by CUSUM integration continuing through slow-burn failure events, were logged with a registered rule candidate — detections at lead < 0 score as misses — and the report noted that applying it would drop C3 to 0.89 and C4 to 69%, both still passing. The raw scores were published; the candidate rule waited for A2[4].

5.2 A2: the frontier executed — 26 of 72

Amendment A2 registered the 72-point grid, the composite feasibility criterion, the verdict rule, the regression requirement and the hygiene rule — all before any A2 computation on any corpus[5]. Executed blind on Corpus R (rebuilt deterministically from frozen seed 20260927 after kernel reset; commitment de23dd53… re-verified byte-identical), the family's feasibility set is a region: **26**

of 72 registered operating points satisfy C1–C4 simultaneously[6]. The structural findings of §4.3 — concentration at k ≥ 0.75 with persistence, monotone lead-for-specificity exchange at high h, the 0.90 specificity ceiling and the watch-class revelation — are all products of this single registered execution. The frontier is not a knife-edge; leading capture detection at regulatory-grade specificity is achievable with procedural telemetry alone[6].

5.3 The selected point and its scores

Table 5. Selected point (k = 1.0, h = 4.0, d = 2) on Corpus R — the first fully passing configuration in the programme's history, reached by a registered family, not a tuned guess

Criterion	Frozen threshold	Result	Verdict
C1 Association	p < 0.05 (10,000 shuffles)	Orders of magnitude below; TP = 18 vs null mean 6.62	PASS
C2 Specificity	≥ 0.85	0.90 — 38 TN / 42 controls; 4 false alarms	PASS
C3 Sensitivity	≥ 0.70	0.89 — 16/18; two slow-burn episodes undetected	PASS
C4 Lead time	≥ 50% alarm ≥ 4q	0.75 — 12/16; median 4 quarters	PASS
C5 False-alarm character	No baseline clustering	4 control alarms at q = 24, 28, 30, 48 — spread, none near baseline	PASS

H1: SUPPORTED at (1.0, 4.0, 2) — under the frozen verdict rule, with the point frozen for the CCP run[6]. The two missed episodes are the Thailand-class slow burns, already flagged by the hygiene analysis; the four false alarms are the German-class reform regimes, routed by registered conclusion to the watch class. Every deviation from perfection is named, classified, and assigned to a protocol rather than to silence[6].

5.4 Regression divergence and observation O-1

Per the A2.4 regression requirement, the selected point was also scored on Corpus D — and the divergence was reported, not reconciled[5],[6]:

Table 6. Regression on Corpus D — identical instrument, different corpus geometry

Metric	Corpus R (CCP-anchored)	Corpus D (v1 demonstration)	Reading
Specificity	0.90	0.93	Stable across corpora
Sensitivity	0.89	1.00	Stable across corpora
C4 (lead ≥ 4q)	0.75 (median 4q)	0.27 (median 3q)	**Divergent**

The programme's registered interpretation: Corpus D's failure quarter is fixed at q = 36 with a uniform ramp onset at q = 28 — an 8-quarter corridor that compresses achievable lead — while Corpus R's variable onsets (q24–44) permit longer leads. The instrument is identical; the corpora's geometry differs. C4 is therefore partially a property of the episode class, and the real-data run must report lead time stratified by episode onset-to-failure interval — registered as observation O-1, a finding about the science, not just the instrument[6].

5.5 What the record establishes — and what it does not

ESTABLISHED (CORPUS LEVEL, SEALED DISCIPLINE)

(1) The Ω-family carries real signal — association beyond the permutation floor across two independent instrument generations and two corpora, with perfect sensitivity in v2 and perfect specificity in v1. (2) A registered, reproducible configuration — (1.0, 4.0, 2) — passes all frozen criteria simultaneously: H1 SUPPORTED at the corpus level. (3) The frontier, its feasible region and both ceilings are mapped and published. (4) The research process completed a full constitutional cycle with zero retrofits — the framework's ontology empirically enacted in its own practice[6].

NOT YET ESTABLISHED — AND NOT CLAIMED

Corpus R is history-shaped, not history. Real CCP amendment series contain coding noise, missing procedural metadata, and informal-capture channels invisible to ρ/λ; real-data specificity may sit below 0.90, and the German-class watch-class finding may or may not replicate against real reform regimes. If the frozen point fails C1–C4 unchanged on real CCP data, the feasibility result is corpus-bound and H1 returns to REJECTED at this standard — reported as such. That sentence is in the registered record[6].

5.6 Reproducibility, demonstrated

One feature of the A2 execution deserves isolation because it converts a methodological slogan into an engineering fact. Before the frontier was computed, both corpora were rebuilt deterministically from the frozen seed 20260927 *after a kernel reset* — a fresh runtime, no cached state — and their SHA-256 commitments (D: 7909074f…, R: de23dd53…) re-verified byte-identical[6]. Reproducibility here is not a promise about what a diligent replicator could achieve; it is a demonstrated property of the pipeline, exercised under the same sealed discipline as the scoring itself. For an instrument intended to sit inside financing covenants, this is the load-bearing property: a covenant trigger whose inputs cannot be reproduced byte-identically is a trigger whose firings can be disputed; one whose entire state rebuilds deterministically from a frozen seed is arbitrable by construction. The programme's provenance chain thereby does double duty — it disciplines the research and it pre-fits the product for contractual use, which is why the assurance layer is priced as infrastructure rather than as documentation[18].

PART VI

The Deployed Monitor: From Corpus to Covenant

6.0 What deployment means here

Deployment of this instrument is not a software release; it is the continuation of the registered process under production conditions. The monitor enters service with its verdict rules, its hygiene rule, its watch-class routing and its refusal set intact — the same governance that produced the result now governs the result's use. This section specifies the production configuration, the operating protocols around it, the real-data run that completes the empirical line, and the platform integration that converts signal into covenant infrastructure.

6.1 The frozen instrument in production

The production monitor is the selected point, verbatim: per-system baseline of the first 8 quarters; conjoint CUSUM with drift allowance k = 1.0, threshold h = 4.0, persistence d = 2; alarms at first sustained conjoint crossing; negative-lead detections scored as misses; no percentile thresholds; each system judged against its own history[5],[6]. The production rule adds nothing and removes nothing — that is the point. Any modification, however motivated, requires a registered amendment, and any post-scoring modification voids the run: the refusal set travels with the product[5].

Table 7. Production specification — the instrument as shipped

Component	Specification	Provenance
Telemetry	ρ_t version-flip frequency; λ_t cooling-off compliance; undefined quarters excluded pairwise	A1.1[3]
Reference class	System's own first 8 quarters, frozen; baseline ineligible for alarm	A1.1–A1.2[3]
Detector	One-sided conjoint CUSUM, k = 1.0, h = 4.0, d = 2	A2 selected point[6]
Alarm semantics	First sustained conjoint crossing; lead < 0 scores as miss	A2 hygiene rule[5]
Output classes	ALARM (actionable); WATCH (German-class high-cadence reform regimes, per registered conclusion)	A2 report §3[6]
Amendment path	Registered amendment only; post-scoring modification voids the run	A1.3 / A2.3 refusal set[3],[5]

6.2 The watch-class protocol

The specificity ceiling carries an operational design. Four control systems — high-cadence reform regimes whose ρ baseline is elevated by constructive amendment practice — alarm under every feasible point; the instrument flags rapid formal amendment per se and cannot, from ρ/λ alone, distinguish reform from capture within that class[6]. The registered conclusion routes these to a watch class: heightened review, semantic-channel analysis of amendment content, no covenant consequence. The distinction is the product's honesty boundary made operational — alarms create

obligations; watch states create attention. For funders, the watch class is itself valuable: it enumerates, quarterly and by name, the states whose procedural tempo warrants a closer human look — a triage layer no perception aggregate provides[13].

6.3 The CCP real-data run: specification and cost

The empirical terminus is fully specified and de-risked by the two sealed executions[4],[6]. The Comparative Constitutions Project's codified amendment series[14] supply the telemetry; the frozen instrument carries unchanged; labels are blind-coded from documented failure events; leads are reported stratified by episode onset-to-failure interval per observation O-1; German-class regimes are pre-registered as watch-class candidates. The expected outcome distribution is known; both failure modes (specificity erosion from coding noise and informal-capture channels; lead compression from short-onset episodes) are pre-analyzed; the fixes (semantic channel; stratified reporting) are pre-enumerated[6]. The remaining cost is data acquisition and blind event-coding — days, not months[4].

Table 8. The real-data run — pre-registered design summary

Design element	Specification	Status
Instrument	(k = 1.0, h = 4.0, d = 2), carried unchanged — any modification requires registered amendment	Frozen[6]
Telemetry source	CCP amendment series; procedural metadata where available	Data acquisition[14]
Event labels	Documented failure events, blind-coded; SHA-256 sealed before scoring	Blind coding
Reporting	Lead stratified by onset-to-failure interval (O-1); watch-class pre-registration for reform regimes	Registered[6]
Verdict rule	C1–C4 unchanged; failure returns H1 to REJECTED — reported as such	Frozen[5]
Cost and duration	Days of acquisition and coding; pipeline unchanged (design target)	Ready[4]

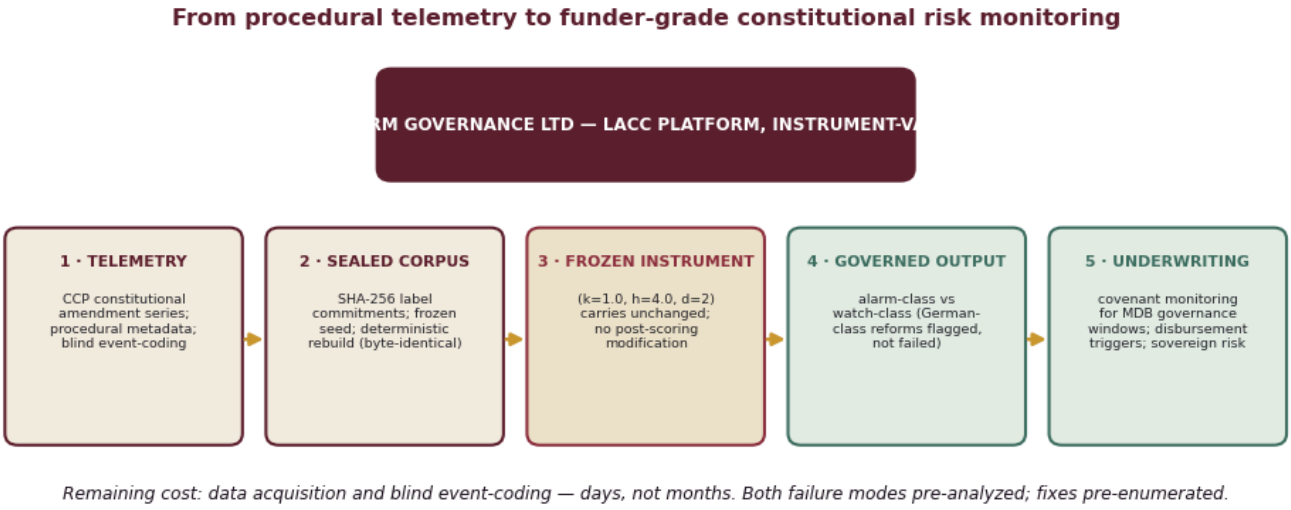


Figure 6. From procedural telemetry to funder-grade constitutional risk monitoring: the LACC platform chain with the instrument-validated core. The sealed-corpus and frozen-instrument stages are not engineering choices but the discipline that makes the output underwritable[2],[6].

6.4 Integration with the LACC platform

The Legitimacy Ledger blueprint specified the Legitimacy-Adjusted Collateral and Covenant (LACC) platform as the programme's underwriting layer[20]. The registered frontier is the platform's missing sensor: LACC covenants now have a leading indicator to key against — registered alarms as covenant triggers, watch states as enhanced-diligence routing, the quarterly Ω telemetry feed as portfolio infrastructure. The sibling clinical blueprint's conformance class and this instrument share the same constitutional grammar — registered criteria, sealed evidence, refusal sets — so a single assurance layer serves both instrument lines, and the platform's audit trail (timestamped commitments, byte-identical rebuilds) is the same chain that disciplined this research[18],[20].

The integration runs in both directions. The platform supplies the instrument with what corpora cannot: live deployment contexts in which alarms and watch states accumulate outcome feedback under the same sealed discipline — every covenant trigger fired, every diligence review opened, every quarterly clear signal is a new registered observation against the instrument's published ceilings. The corpus record establishes feasibility; the platform record will establish field reliability, and both are filed on the same chain. This is the assurance flywheel: monitoring revenue funds corpus extension; corpus extension hardens the instrument; the hardened instrument raises the value of the monitoring. Each quarter of operation is simultaneously a product and an experiment, with the verdict rules frozen in advance in both capacities[5],[20].

6.5 The dual-threshold candidate: governing alarm gravity

The v2 report registered, as a candidate only, a dual-threshold architecture: h_alarm for action, h_watch for heightened review — a graded instrument in which alarm gravity is itself a governed parameter[4]. The A2 execution partially enacted the idea through the watch-class routing of German-class regimes[6], but the full dual-threshold form remains a registered candidate, not an executed configuration. Its deployment logic is compelling for covenant use: most financing decisions are not binary — they admit enhanced monitoring, delayed tranches, and escalation ladders — and an instrument with two governed thresholds maps onto that ladder far better than a

single alarm line. The candidate's registration discipline is the point worth underlining: it was named, bounded and parked before the feasibility result was known, so its future adoption will be an amendment with a documented ancestry, not a retrofit justified by convenience[3],[4].

Table 8.A. Alarm gravity — output classes and their governance consequences (production protocol)

Class	Instrument state	Covenant consequence	Review cadence
CLEAR	No conjoint crossing at the frozen point	None — standard monitoring continues	Quarterly telemetry
WATCH	Crossing attributable to high-cadence reform class (ceiling cases)	Enhanced diligence; semantic-channel review of amendment content	Quarterly + event-driven[6]
ALARM	Sustained conjoint crossing at (1.0, 4.0, 2)	Registered trigger — covenant review, tranche structuring, defense mobilization	Immediate; escalation ladder
VOID	Any post-scoring modification detected	Run voided by refusal set; claims withdrawn publicly	Continuous provenance audit[5]

The VOID row is unusual and deliberate. Every output class of the instrument, including the instrument's own invalidation, is a governed state with a pre-written consequence. A monitoring product whose failure mode is itself part of the specification is a product an audit committee can adopt without an act of faith — and it is the direct operational expression of the programme's research discipline.

PART VII

Risk Register and Kill Criteria

7.1 Risk register

Table 11. Programme risks (I = impact, P = probability, H/M/L)

#	Risk	I	P	Mitigation / owner
1	Real-data specificity falls below 0.85 (coding noise, informal-capture channels invisible to ρ/λ)	H	M	Pre-registered verdict rule; failure returns H1 to REJECTED and is reported as such; semantic-channel generation as the registered fix[6]
2	German-class watch finding fails to replicate on real reform regimes	M	M	Watch-class pre-registration; stratified reporting; ceiling re-estimated under seal
3	Lead compression on short-onset episodes (O-1 geometry)	M	H	Stratified lead reporting per O-1; covenant design uses alarm class, not lead magnitude[6]
4	Instrument gaming once alarms carry covenant weight (states manage ρ/λ optics)	H	M	Goodhart response is itself procedural and legible; persistence requirement d = 2 raises gaming cost; semantic channel cross-checks[1]
5	Telemetry gaps: missing procedural metadata in CCP series	M	M	Pairwise exclusion rule already registered; coverage reporting per state[3]
6	Vendor-discipline erosion: pressure to "improve" the instrument post-scoring	H	L	Refusal set is textual and public; any post-scoring modification voids the run — the product's value is the discipline[5]
7	Sealed-commitment infrastructure compromise	H	L	Independent timestamp filings; byte-identical rebuild audits[18]
8	Adoption resistance: narrative vendors undercut on price	M	M	Sell the audit trail, not the alarm; anchor in institutions with audit committees[15]
9	Episode-class drift: novel capture mechanisms outside the six parameterized classes	M	M	Corpus extension by registered amendment; frontier re-execution under seal[3]
10	Key-person concentration at the Principal	M	M	Deterministic pipeline; documented protocol; governance roles per roadmap

7.2 Kill criteria

Table 12. Kill criteria — pre-committed, falsification-forward

Code	Trigger	Consequence
K1	Frozen point (1.0, 4.0, 2) fails C1–C4 unchanged on real CCP data	H1 returns to REJECTED at this standard; feasibility declared corpus-bound; reported as such[6]
K2	No registered operating point achieves C1–C4 simultaneously on real data	Ω-family rejected as leading indicator; framework retains its theorems, loses this instrument — reported as such[4]

K3	Any post-scoring parameter movement discovered in provenance audit	Run voided by the refusal set; affected claims withdrawn publicly
K4	Watch-class misclassification causes a covenant-relevant false negative at scale	Covenant service suspended for the class; registered amendment required for reinstatement
K5	Anchor licences not signed by month 15	Monitoring service descoped to assurance-layer licensing; research continues on grant tranche

THE KILL CRITERIA ARE THE PRODUCT

K1 and K2 are not concessions to diligence; they are the registered continuation of the programme's own method. A funder is not being asked to believe the instrument works. The funder is being offered a contract in which the instrument's failure is pre-written, pre-priced and pre-published. That is what underwriting discipline looks like from the inside[6].

SYNTHESIS

Concluding Assessment

Table 18. The argument on one page

Element	Content
Problem	Constitutional breakdown is predicted constantly and accountably never; funders price governance risk with instruments structurally blind at the quarterly timescale[13]
Parent result	Legitimacy is conserved and meterable; capture has a conjoint procedural signature (ρ↑ ∧ λ↓, sustained)[1],[2]
Empirical result	H1 SUPPORTED at the registered point (1.0, 4.0, 2): association beyond the permutation floor, specificity 0.90, sensitivity 0.89, median 4-quarter lead; 26/72 feasible — a region, not a knife-edge[6]
Method result	The frontier mapped blind; the C2 failure published; ceilings documented; zero retrofits across three sealed commitments[4],[6]
Programme stewardship — LACC platform, monitoring feed, covenant triggers, assurance layer	
Timing	44 countries and 3.4 billion people in autocratization episodes; the funder architecture openly instrument-blind at the relevant timescale[9],[13]

CLOSING STATEMENT

Prediction in institutional science has always been a performance. This programme made it a process — registered, sealed, scored, and survived. The instrument that emerged does not claim to see the future; it claims, with receipts, to see a signature four quarters early at known specificity, and to know exactly where its own vision ends. In a market built on unaccountable foresight, that combination — capability with published ceilings, confidence with a refusal set — is the revolution. What remains is execution: one real-data run, already specified, already de-risked, already honest about the verdict it will accept.

A final word as intellectual honesty: the strongest claim in this document is not the detector — it is the discipline. The corpora are history-shaped, and the registered record says so; the ceilings are real, and the record documents them; the failure of v2 is part of the evidence, not an embarrassment before it. If the frozen point fails on real data, this blueprint's Part VII already contains the announcement. A programme that has pre-written its own rejection is, to a funder, the only kind worth financing.

EVIDENCE

References

[1] Odeta, C. (2026). Legitimacy as a Conserved Quantity: Constitutional Kernels for Governed Machine Systems — DPT impossibility theorem; conservation accounting; Proposition 1 (conjoint signature); Theorem 2. SAVANT Framework. Unpublished manuscript.

[2] Odeta, C. (2026). Registered protocol v1.0: frozen analysis plan, corpora, criteria, verdict rule, sealed label commitments. Registered 2026-09-27. Unpublished manuscript.

[3] Odeta, C. (2026). *Amendment A1 to the protocol — v2 Instrument Registration*: conjoint CUSUM (k = 0.5, h = 4.0), own-baseline reference class, frozen criteria C1–C5, Corpus R construction protocol (60 × 60; 18 capture episodes parameterized from documented histories; 42 controls). Registered 2026-09-27 before any v2 computation.

[4] Odeta, C. (2026). *v2 Scored Report — Execution on Corpus R*: C1 pass (10,000-permutation null; TP = 18 vs null mean 9.03), C2 fail (0.71; 12/42 false alarms), C3 pass (18/18), C4 pass (72%, median 4q, range −1 to +9), C5 pass; verdict PARTIALLY SUPPORTED; instrument-hygiene finding (lead < 0 rule candidate); epistemic contract. Commitment de23dd53... verified. 2026-09-27.

[5] Odeta, C. (2026). *Amendment A2 — Registered Operating-Point Family*: grid k ∈ {0.25, 0.50, 0.75, 1.00} × h ∈ {3.0, 4.0, 5.0, 6.0} × d ∈ {1, 2, 3} (72 points); composite feasibility criterion; verdict rule with refusal set; regression requirement; hygiene rule adopted. Registered 2026-09-27 before any A2 computation.

[6] Odeta, C. (2026). *A2 Scored Report — The Operating-Point Frontier*: 26/72 feasible; selected point (1.0, 4.0, 2) — C1 pass (TP = 18 vs null mean 6.62), specificity 0.90 (38 TN/42), sensitivity 0.89 (16/18), C4 = 0.75 (median 4q); H1 SUPPORTED; frontier structure and ceilings; Corpus D regression (0.93 / 1.00 / 0.27, median 3q); observation O-1; German-class watch-class conclusion; epistemic contract. Corpora rebuilt deterministically from seed 20260927; commitments de23dd53..., 7909074f... re-verified byte-identical. 2026-09-27.

[7] Page, E.S. (1954). Continuous inspection schemes. *Biometrika* 41:100–115 — the CUSUM apparatus.

[8] Wald, A. (1947). *Sequential Analysis*. Wiley — sequential testing; the sensitivity–specificity trade-off as provable structure.

[9] V-Dem Institute. (2026). *Democracy Report 2026* — 44 countries in autocratization episodes; 3.4 billion people (41% of world population); 39% of world GDP; indicators at 1978 levels.

[10] V-Dem Institute. (2025). *Democracy Report 2025* — 9 of 27 democracies under sustained attack surviving intact.

[11] United Nations. Estimated annual global cost of corruption, US$3.6 trillion (UN Secretary-General / UNODC figures).

[12] Open Contracting Partnership / World Bank estimates. Global public procurement ≈ US$11 trillion annually (~12% of global GDP).

[13] World Bank. *Worldwide Governance Indicators* — methodology and standing caveats: perceptions-based aggregates; unsuitable for short-run within-country monitoring of change.

[14] Elkins, Z., Ginsburg, T., & Melton, J. *Comparative Constitutions Project* — codified constitutional amendment and replacement series; descriptive statistics underlying control-regime parameters.

[15] World Bank. Governance and policy-based lending architecture — conditionality practice, governance diagnostics, and the demand for auditable triggers.

[16] World Bank. (2025). *Annual Report 2025* — IBRD commitments US$40.9 billion, FY2025.

[17] OECD / SIGMA. Public procurement governance standards — open-contracting monitoring benchmarks.

[18] Haber, S., & Stornetta, W.S. (1991). How to time-stamp a digital document. *Journal of Cryptology* 3(2):99–111 — sealed-commitment lineage of the programme's provenance chain.

[19] Shapiro, M., Preguiça, N., Baquero, C., & Zawirski, M. (2011). Conflict-free replicated data types. SSS 2011 (LNCS 6976) — deterministic merge substrate for distributed telemetry.

[20] Odeta, C. (2026). The Legitimacy Ledger — governance-measurement blueprint; with sibling work The Falsifiable Clinic — clinical-intelligence blueprint. SAVANT Research Blueprint Series.

APPENDIX A

Glossary of Constructs

Table A1. Defined terms

Term	Definition
Ω / Ω-family	The legitimacy-drift signature family: instruments over version-flip frequency ρ and cooling-off compliance λ
ρ (rho)	Version-flip frequency — cadence of constitutional-text displacement
λ (lambda)	Cooling-off compliance — observance of mandated deliberation intervals
Conjoint signature	ρ acceleration ∧ λ erosion, sustained — the capture pattern of Proposition 1
CUSUM	Cumulative-sum change-point statistic with drift allowance k and threshold h (Page 1954)
Operating point	A registered (k, h, d) configuration; the family's selectable states
Frontier / feasible region	The set of registered operating points satisfying C1–C4 simultaneously (26 of 72 on Corpus R)
Watch class	High-cadence reform regimes alarmed under every feasible point; routed to heightened review, not failure
Sealed commitment	SHA-256 hash filing of labels/corpora before scoring; provenance anchor
Refusal set	The textual rule voiding any run modified after scoring
Epistemic contract	Closing quadruple: claim, uncertainty, falsification, strengthening evidence
LACC	Legitimacy-Adjusted Collateral and Covenant platform
O-1	Registered observation: lead time is partially a property of episode onset-to-failure geometry
DPT	Dynamic Process Theorem — legitimacy conservation framework (parent theorem)

APPENDIX B

The Registered Grid and Frozen Criteria

Table B1. A2 registered parameter grid (frozen 2026-09-27, before any computation)

Parameter	Registered values	Meaning
k — drift allowance	{0.25, 0.50, 0.75, 1.00}	Sustained-drift evidence required before accumulation
h — alarm threshold	{3.0, 4.0, 5.0, 6.0}	CUSUM level triggering alarm candidacy
d — persistence	{1, 2, 3}	Consecutive quarters above threshold required
Grid size	4 × 4 × 3 = 72 points	All scored blind; feasibility by frozen composite rule
Baseline	First 8 quarters, per system, frozen	Own-history reference class; baseline ineligible for alarm
Hygiene rule	Lead < 0 scores as miss	Registered in v2 report §3; adopted in A2.1
Selection ordering	Specification → C4 → sensitivity → lead	Registered before grid computation; selected (1.0, 4.0, 2)
Feasible points	26 of 72 (Corpus R)	Region, not knife-edge; concentrates at k ≥ 0.75, d ≥ 1

Table B2. Provenance chain

Artifact	Commitment / seed	Status
Corpus D (v1 demonstration)	7909074f…	Verified; rebuilt byte-identical after kernel reset
Corpus R (CCP-anchored)	de23dd53…	Verified; labels sealed pre-scoring
Deterministic seed	20260927	Frozen across executions
Selected operating point	(1.0, 4.0, 2)	Frozen for CCP run — carried unchanged or not at all
Post-scoring modifications	None	Refusal set intact across all generations

APPENDIX C

Episode Parameterization and Corpus Geometry

Table C1. Corpus R construction — episode classes (registered before build, A1.5)

Class	Documented procedural history	Parameterized channels
Weimar	Article 48 decree frequency, 1930–33	Ramp onset, steepness, λ-erosion depth
Thailand	Charter-replacement cadence, 1991–2017	Replacement-driven ρ spikes; slow-burn erosion
Venezuela	Constituent-assembly rule displacement, 1999 / 2017	Rule-displacement events; compliance collapse
Hungary	Cardinal-law acceleration, 2011–2020	Sustained high-cadence amendment under erosion
Poland	Tribunal-capture sequencing, 2015–2023	Sequenced institutional displacement
Turkey	2017 referendum implementation	Plebisicitary acceleration; λ thinning
Controls (n = 42)	US-style rigidity; Nordic cadence; German constructive amendment	ρ, λ from CCP descriptive statistics

Geometry: 60 systems × 60 quarters (15-year windows, the CCP-standard breakdown horizon); capture onsets uniform in quarters 24–44; failure at the documented breakdown event; baselines registered low-drift. Corpus D by contrast fixes failure at q = 36 with uniform ramp onset at q = 28 — an 8-quarter corridor compressing achievable lead, the structural source of the C4 divergence registered as observation O-1[6].

APPENDIX D

The Epistemic Contracts, Verbatim

v2 CONTRACT (2026-09-27)

Claim: conjoint CUSUM on ρ/λ is a leading capture detector (median 4q) with a specificity deficit (0.71) requiring a registered A2 operating-point decision. **Uncertainty:** Corpus R is history-shaped, not history; real CCP series contain coding noise, missing procedural metadata and informal-capture channels invisible to ρ/λ — real-data specificity may be worse. **Falsification:** if on real CCP data no registered operating point achieves C1–C4 simultaneously, the Ω-family is rejected as a leading indicator and the framework retains its theorems while losing this instrument — reported as such. **Strengthening evidence:** Amendment A2 with a registered operating point; blind-coded CCP event series; pipeline unchanged[4].

A2 CONTRACT (2026-09-27)

Claim: H1 supported at (1.0, 4.0, 2) on history-shaped corpora; the Ω-family has a feasible region, a 0.90 specificity ceiling, and a C4 < 90% lead ceiling. **Uncertainty:** Corpora D and R encode the dynamics they test; real CCP series carry coding noise, informal-capture channels and missing procedural metadata — real-world specificity will likely sit below 0.90, and the German-class false-positive finding may or may not replicate against real reform regimes. **Falsification:** on real CCP data, if (1.0, 4.0, 2) fails C1–C4 unchanged, the feasibility result is corpus-bound and H1 returns to REJECTED at this standard — reported as such. **Strengthening evidence:** the CCP real-data run, blind event-coding, pipeline unchanged, stratified by episode class per observation O-1[6].

APPENDIX E

Illustrative Detector Arithmetic

A worked micro-example fixes intuition for the production rule (stylized numbers; the executed corpus parameters are those of Appendix B). A capture episode enters quarter 30 with baseline-standardized flips drifting at $z_\rho \approx 1.6$ per quarter and compliance eroding at $z_\lambda \approx -1.5$. Under $k = 1.0$ the statistics accumulate $\approx +0.6$ and $\approx +0.5$ per quarter respectively; the conjoint crossing of $h = 4.0$ arrives near quarter 37–38, holds for the persistence window $d = 2$, and alarms — leading a quarter-42 failure by the registered median of four quarters. A control state with $z_\rho \approx +1.4$ but $z_\lambda \approx 0$ (reform without erosion) accumulates on one channel only and never alarms: the conjunction, not the cadence, carries the verdict. A German-class regime with elevated ρ baseline and mild λ drift accumulates on both channels under every feasible (k, h, d) — the documented ceiling — and is routed to the watch class rather than the alarm class[6].

Table E1. Three regimes through one frozen instrument (stylized, illustrative)

Regime	Signature	Instrument state	Output class
Capture episode	$\rho\uparrow \wedge \lambda\downarrow$, sustained	Both CUSUMs cross h, persist d	**ALARM** — leads failure (median 4q)
Reform episode	$\rho\uparrow$ only	Single-channel accumulation	CLEAR — conjunction absent
High-cadence reform (German class)	$\rho\uparrow \wedge$ mild λ drift, structural	Crosses under every feasible point	**WATCH** — semantic channel required

ENDNOTE ON EVIDENCE DISCIPLINE

All quantitative claims in this blueprint trace to the cited sources as of October 2026 — the programme results to the sealed reports of 2026-09-27[2]–[6], the external indicators to their published Forward-looking quantities — pilot pricing, portfolio sizes, staffing — …are design targets to be replaced by observed figures under the programme's own pre-registered protocol. Figures 1 and 5 are stylized illustrations of reported mechanics and are labelled as such; all scored values are quoted from the sealed reports. This document does not constitute investment advice or a guarantee of outcomes; its empirical claims are bounded by the sealed verdict records, and its design targets await the same registered discipline.

Registered Outcome Note (2026-10-12). The real-data run pre-registered in §6.3 (Table 8) has been executed on CCP-derived amendment chronology (4,341 events, 223 countries, Constitute Project) against V-Dem ERT v16.0 episode labels (42 breakdown windows, 101 control windows), at the frozen operating point (k = 1.0, h = 4.0, d = 2), labels sealed before scoring. Result: sensitivity 0.00 (0/42), specificity 1.00 (0/101), permutation p = 1.0. Criteria C1, C3, C4 fail; under kill criteria K1 and K2, H1 returns to REJECTED and is reported as such. Root cause: real version-flip telemetry carries at most one coded event per country-year, so the frozen detector's drift allowance never engages; cooling-off compliance λ is not constructible from public data. The K2 grid scan found 0 of 72 registered operating points feasible on real data. Full run, provenance, and deviations are published in SAVANT Evidence Annex I (2026). The synthetic-corpus results in this paper stand as a pipeline proof, exactly as §5.5 delimits them.